Compact size 10/100M Industrial PoE Media Converter

LT-IMC30FP-E Series



Features:

- 1*10/100Base-TX RJ45 PoE port, 1*100Base-FX SFP(/SC/ST/FC optional) fiber port
- DC 44~57V input, redundant power supply with polarity reverse protection
- Complies with IEEE802.3af PoE and IEEE802.3at PoE+ standard
- Powerful Dip switch function:

  1.LFP; 2. Flow control; 3. 250m long distance; 4. PoE watchdog
- Support 10K Bytes Jumbo frame and 1M large buffer
- Support 4KV surge protection and ESD: Air-15kV, Contact-8kV Protection
- IP40 fan-less and Din-rail hardware design with compact size(30*85*85mm)
- Operation temperature: -40 ℃ ~+75℃

Overview

The SmartByte LT-IMC30FP-E series are unmanaged industrial grade media converter with 1-port 10/100Base-TX PoE and 1-port 100Base-FX fiber optical interface. They comply with IEEE802.3af, IEEE802.3at standard PoE protocol, the Max power consumption can reach 30W (PoE+) Moreover, they are featuring with LFP, flow control, 250m long distance transmission mode and PoE watchdog function, which all can be configured by the Dip switch on the top panel.

LT-IMC30FP-E series are also high cost-effective easy-to-use devices, which provide essential industrial Ethernet networking function, such as wide range power input 44-57VDC, redundant power design with polarity reverse protection, robust IP40 fan-less compact size housing with Din-rail installation, wide operation temperature from -40℃ ~+75℃ as well as high-level EMI/EMC capability and so on. It is the best choice for heavy industrial factory, transportation, oil & gas, chemical, IP Surveillance and processing automation area where environmental conditions exceed commercial product specifications

Technical specification

<table>
<tr><td>Model No.</td><td colspan="3">LT-IMC30FP-E</td></tr>
<tr><td rowspan="2">Interface</td><td>Fiber port</td><td colspan="2">Copper RJ45 port</td></tr>
<tr><td>1</td><td colspan="2">1</td></tr>
<tr><td>Ethernet</td><td colspan="3">1*10/100Base-TX PoE
1*100Base-FX SFP (SC/ST/FC optional)</td></tr>
<tr><td>Standard</td><td colspan="3">IEEE 802.3 10Base-T Ethernet
IEEE 802.3u Fast Ethernet
IEEE 802.3x Full-Duplex Flow Control
IEEE 802.3af Power-Over-Ethernet
IEEE 802.3at Power Over Ethernet plus PSE</td></tr>
<tr><td>Dip Switch</td><td colspan="3">1. LFP
2. Flow control
3. 250m long distance mode
4. PoE watchdog</td></tr>
<tr><td rowspan="6">LED Indicators</td><td rowspan="2">P(Power indicator) Green</td><td colspan="2">Off: the device is power off or failed</td></tr>
<tr><td colspan="2">On: the device power on is normal</td></tr>
<tr><td rowspan="3">S(PoE usage indicator) Red</td><td colspan="2">Off: total PoE usage < rated 50%</td></tr>
<tr><td colspan="2">Blinking: rated 50% < total PoE usage < rated 90%</td></tr>
<tr><td colspan="2">On: total PoE usage≥ rated 90%</td></tr>
<tr><td rowspan="2">1 (Copper port)</td><td>Green indicators</td><td>Yellow indicators</td></tr>
<tr><td>Off: ports link down</td><td>Off: PoE not working</td></tr>
</table>

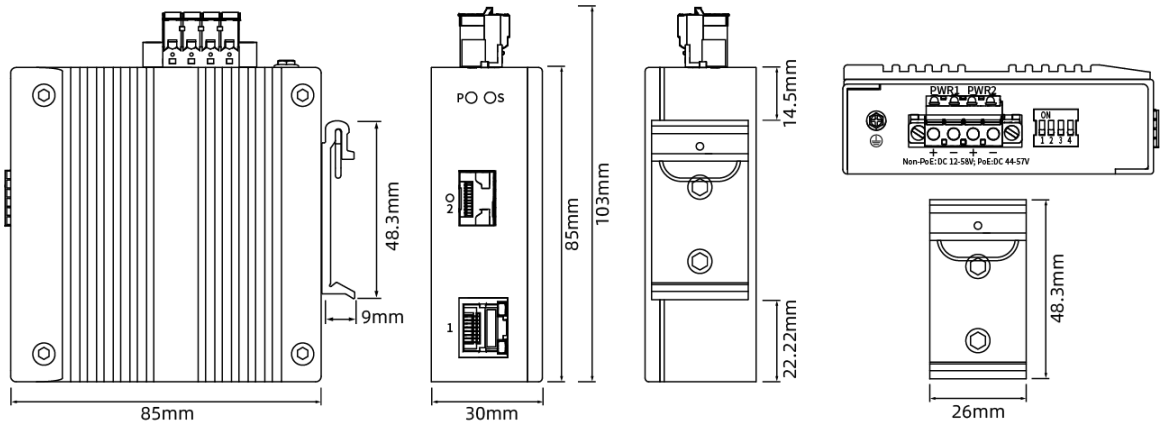
		On: ports link up	On: PoE working
		Blinking: data on TX/RX	
	2 (Fiber port) Green	Off: ports link down	
		On: ports link up	
		Blinking: data on TX/RX	

Power parameters	
Input voltage	44-57VDC, redundant power input
Input current	0.75A Max
Total power consumption	Full loading without PoE ≤ 2W PoE power budget ≤ 30W
Connector	Removable 4-pin terminal block
Reverse polarity protection	Support
Switching features	
Switching capacity	0.4G
Packet forwarding rate	0.59Mpps
MAC address table	8K
VLAN	Support the transmission of the packets with VLAN tags ranging from 1 to 4096
Buffer	1M
Forwarding delay	<10us
Jumbo Frame	Support 10Kbytes
MDI/MDIX	Support
Watchdog	Support
Mechanical structure	
Case protection	IP40
Installation method	Din-rail
Dimension(W*D*H)mm	30*85*85mm
Weight	0.2 kg
Operating environment	
Operating temperature	-40℃~+75℃
Storage/transportation temperature	-40℃~+85℃

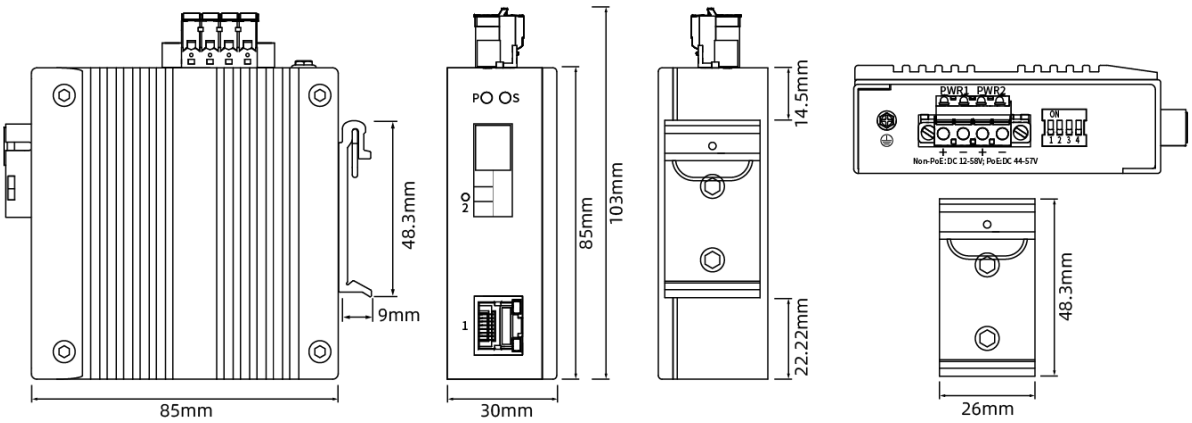
Relative humidity	5%~95%（non-condensing）
Industrial Standard	Surge protection of power: IEC 61000-4-5 Level 3 (4KV/2KV) （8/20us）
	Surge protection of Ethernet ports: IEC 61000-4-5 Level 3 (4KV/2KV) （10/700us）
	DIP: IEC 61000-4-11 Level 3（10V）
	ESD: IEC 61000-4-2 Level 4（8K/15K）
	Shock：IEC 60068-2-27
	Free fall：IEC 60068-2-32
	Vibration：IEC 60068-2-6
Certification	CCC/CE/FCC/RoHS/UKCA
MTBF	> 200,000 hours
Warranty	5 years

Structure diagram

SFP model



SC/ST/FC model



Order information

Model	Description
LT-IMC30FP-E	10/100Base-TX to 100Base-FX unmanaged compact size industrial PoE media converter with 1*10/100Base-TX RJ45 PoE port and 1*100Base-FX SFP slot, DC44-57V input, redundant dual power supply, Din-rail installation. Fiber port transmission distance depending on the SFP module; Operation temperature: -40℃～+75℃
LT-IMC30FP-SC/FC /ST-E	10/100Base-TX to 100Base-FX unmanaged compact size industrial PoE media converter with 1*10/100Base-TX RJ45 PoE port and 1*100Base-FX SC/FC/ST port, DC44-57V input, redundant dual power supply, Din-rail installation; Multi-mode/Single mode/ dual fiber/single fiber, 2KM -120KM optional; Operation temperature: -40℃～+75℃